Jakob Shurtleff

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EDUCATION

B.S. in Physics August 2023 - December 2027
Brigham Young University - Idaho Rexburg, Idaho

- emphasis on computational methods and machine learning applications.

EXPERIENCE

Physics Lab Coordinator & Teaching Assistant February 2026 - Present
BYU-Idaho Physics Department Rexburg, ID

- Coordinated daily operations and logistics for the university physics homework help center, facilitating peer instruction and academic support resources.
- Served as Teaching Assistant for Physics 121 (Principles of Physics I); clarified mechanics concepts, and graded coursework.

Volunteer Physics Tutor August 2025 - February 2026
BYU-Idaho Physics Department Rexburg, ID

- Conducted one-on-one tutoring to help physics students bridge the gap between complex principles and systematic methodologies for solving problems in introductory and intermediate physics classes.

TECHNICAL SKILLS

- Languages & Libraries: Python (NumPy, Polars, Scikit-learn, Matplotlib, pytorch), C++, C#, SQL.
- Developer Tools: Git, Visual Studio Code, Linux, REST APIs.

TECHNICAL PROJECTS

AI projects
Independent Computational & AI Development

- Engineered a Physics Informed Neural Net (PINN) for modeling orbital systems from noisy telemetry data, including building the dataset of 500 example orbits leveraging runge-kutta method.
- Built custom LLM utilities and agential workflows, using pytorch and polars for data and training pipelines and deploying multiple agents dynamically in a custom agential harness.

Nested sampling research April 2026 - Present
Research under Dr. Lance Nelson.

- Utilized HPC resources to perform simulations of hundreds of silicon atoms using Nested Sampling to determine viability of silicon in outperforming current materials.

CERTIFICATIONS

- AI Engineering and Development - Coursera (2025).